



EXECUTIVE SUMMARY

Automated Credential Exploitation of Ubuntu Server

IT312: Cybersecurity Foundations
Using Kali Linux (Metasploit Lab)

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Introduction



Penetration testing [Image Source: Vocal Media].

This report details a simulated penetration test targeting a Linux-based server within the TechLab environment. The objective was to move beyond simple reconnaissance and demonstrate a functional system compromise using the Metasploit Framework. Identifying and weaponizing a common security misconfiguration (weak SSH authentication) in this engagement proves how easily an attacker can pivot from network discovery to full command-line access.

Problem Summary

Modern enterprise security often fails not at the software level, but at the administrative level. The target Ubuntu server was evaluated to determine if its remote management services were susceptible to automated *Credential Guessing*. Without robust password policies or account lockout mechanisms, the server remained a high-risk target for *Initial Access* via brute-force methodologies.

Methodology

For this project, the Metasploit Framework was accessed via the *Kali Linux* terminal (msfconsole). This interface is the industry standard for professional red-teaming because it allows for the integration of a PostgreSQL database to track hosts, services, and discovered vulnerabilities in real-time.

Phase 1: Target Reconnaissance

Initial discovery was performed using the *db_nmap* command. This allowed the scan results to be saved directly into the Metasploit database for follow-up exploitation.

- **Target IP:** 192.168.234.3
- **Identified Vector:** Port **22/tcp (SSH)** was confirmed as *Open*.

- **Significance:** SSH is a high-value target because a successful login grants the attacker an encrypted, stable shell on the victim machine.

```
[*] 192.168.234.3 - SSH session 1 closed. Reason: User exit
msf auxiliary(scanner/ssh/ssh_login) > services
Services
=====
```

host	port	proto	name	state	info	resource	parents
192.168.234.3	22	tcp	ssh	open		{}	
192.168.234.3	80	tcp	http	open		{}	

Fig.1. Nmap Scan Results showing Port 22 and 80 open.

The Anatomy of the Hack

The following steps outline the exploitation analysis. They show how the vulnerability was identified and successfully leveraged to gain unauthorized access:

Phase 2: Attack Configuration

The `auxiliary/scanner/ssh/ssh_login` module was selected. Unlike a manual login attempt, this Metasploit module automates the testing of hundreds of credential pairs per minute.

- **Configuration:** The RHOSTS variable was set to the target's IP, and a specific user list was provided to test for weak authentication.
- **Why it worked:** Many systems use default or easily guessable credentials (like `admin` or `ubuntu`), which are the first things an automated scanner will try.

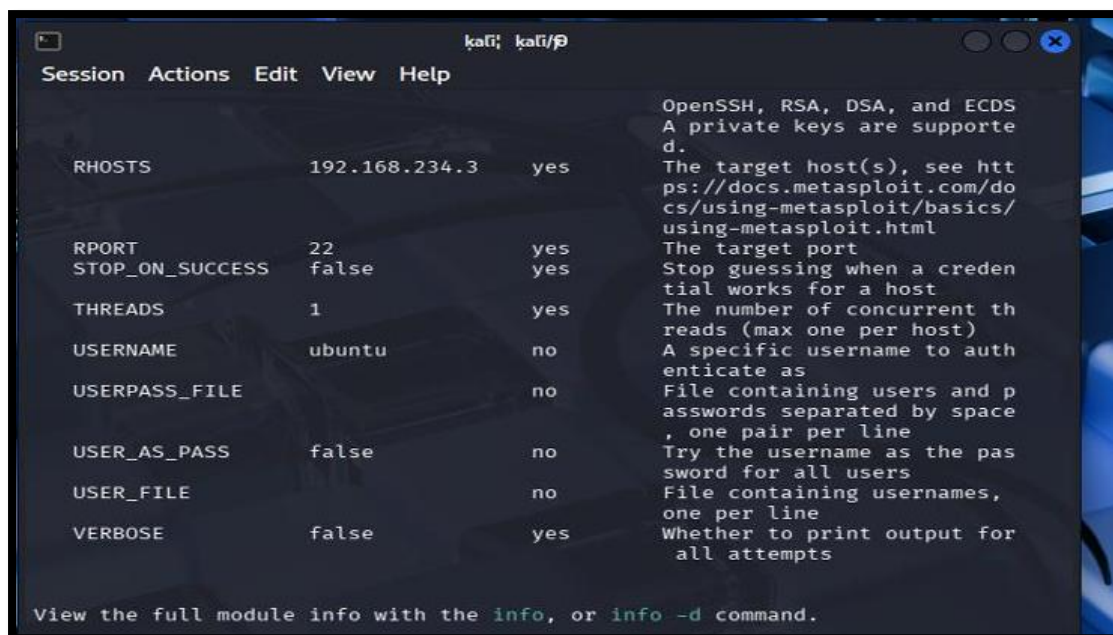


Fig.2. Module Configuration showing RHOSTS 192.168.234.3.

Phase 3: Execution and Exploitation

Upon running the module, Metasploit performed a Brute Force attack.

- **Findings:** The system identified a successful login pair: `ubuntu:ubuntu`.
- **Impact:** A background SSH session (Session 1) was instantly opened, bypassing all perimeter security.

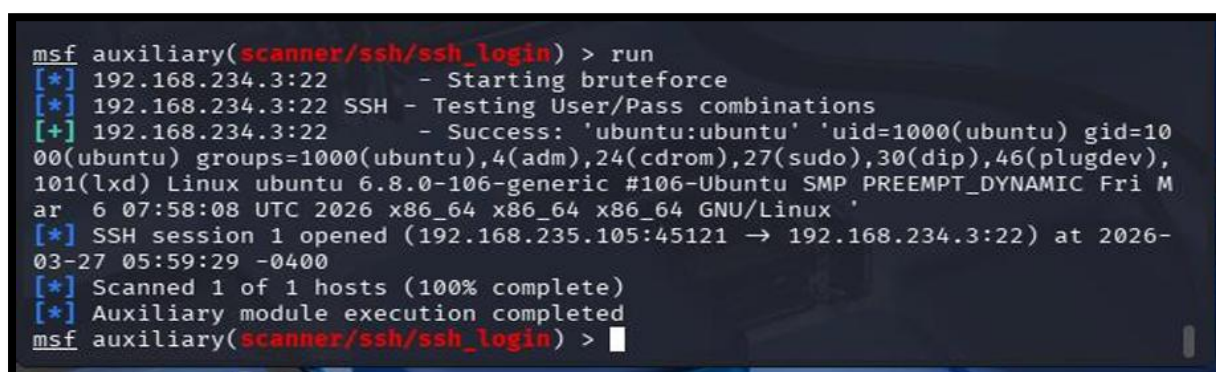


Fig.3. Successful Exploitation showing "Success: 'ubuntu:ubuntu'".

Phase 4: Post-Exploitation (Proof of Access)

To prove the breach was successful, the session was interacted with to execute administrative commands.

- **Command whoami:** Returned *ubuntu*, confirming I had assumed the user's identity.
- **Command hostname:** Confirmed access to execute code directly on the victim's hardware.



The image displays two terminal window screenshots. The top screenshot shows the command 'whoami' being entered, followed by the output 'ubuntu'. The bottom screenshot shows the command 'hostname' being entered, followed by the output 'ubuntu'. Both screenshots have a dark background with light-colored text.

Fig.4. *Proof of Access showing terminal results for whoami.*

Vulnerability Research

The breach was made possible by CWE-521 (Weak Password Requirements). The system failed to provide even basic protection against credential stuffing by allowing the password to match the username. Furthermore, the lack of CWE-307 (Improper Restriction of Excessive Authentication Attempts) allowed Metasploit to continue guessing passwords without being blocked by the system's firewall. This mirrors CVE-1999-0502, a vulnerability class that remains a top-tier risk in modern cloud and local environments.

Conclusion / Recommendations

The reconnaissance and subsequent exploitation prove that the TechLab network's Ubuntu server is highly vulnerable to automated credential attacks. To remediate these issues, the organization should:

1. **Enforce Complex Passwords:** Require 14+ characters with mixed complexity.
2. **Use SSH Keys:** Disable password-based logins entirely in favor of cryptographic keys.

3. **Deploy Fail2Ban:** Implement automated blocking for IPs that fail multiple login attempts.
4. **Enable Multi-Factor Authentication (MFA):** Add a second layer of defense to prevent access even if a password is stolen.